

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	D. Yaswanth Sri Dana Ganesh	Reg. No.:	192425351
Date:		Duration:	60 minutes

Code of Conduct

I D. Yaswanth Sri Dana Ganesh(192425351) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Dana Ganesh

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

1. Introduction

Meteorological and climate-science organisations generate enormous volumes of satellite imagery — infrared, visible and radar — every day to monitor weather systems and long-term climate trends. Much of this data remains locked inside specialised scientific formats and software, limiting its use to trained researchers and leaving policymakers, disaster-management teams and regional planners without timely, accessible insight. The assigned problem statement therefore calls for a web-based Satellite Data Visualization Platform that ingests raw satellite data from public sources such as NASA and NOAA, presents it through interactive maps and time-series charts, detects patterns such as cyclone formation or temperature anomalies, and allows non-expert users to explore, compare and export processed climate information.

This report presents a literature survey of seven recent (2023–2026) research papers relevant to this problem statement. Section 2 reviews each paper individually. Section 3 presents a comparative analysis across the seven studies. Section 4 synthesises the critical gaps in the existing literature relative to the assigned problem, and Section 5 proposes future directions before the report concludes with a full reference list.

2. Selection and Review of Research Papers

Seven papers published between 2023 and 2026 were selected for their direct relevance to the sub-problems named in the assigned brief: web-based satellite/climate data ingestion and access, interactive map- and time-series-based visualization, Web-GIS architectures for hazard management, and AI-based pattern/anomaly detection (cyclone intensity) from satellite imagery.

2.1 SalienTime: User-Driven Selection of Salient Time Steps for Large-Scale Geospatial Data Visualization

Author(s) / Year: Chen, Peng, Huang, Li, Ye, and Wang (2024)

Venue: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24), ACM

Objective: To help users of large satellite/geospatial time-series archives quickly locate the most meaningful ("salient") moments in a dataset instead of scrubbing through a timeline manually.

Methodology: A need-finding study with five Earth-science and visualization experts informed the design of a CNN-based autoencoder combined with a dynamic-programming algorithm that scores time steps on structural difference, statistical variation and temporal spread. The resulting scores drive an interactive web system built with React, D3.js and Mapbox GL JS, offering a map view, a timeline with contextual trend charts, and a latent-space explorer.

Dataset / Tools: NASA Earthdata and NOAA reanalysis datasets (2-metre air temperature, sea-level pressure), WAVEWATCH III wave-height data, and GOES-16 Cloud and Moisture Imagery used for a hurricane case study.

Key Findings: The proposed selection method produced lower reconstruction error (RMSE/SSIM) than uniform or arc-based sampling baselines, particularly for small numbers of selected frames, and scored 85.5 on an adapted System Usability Scale versus 62.5–77.5 for three existing satellite-data portals (NASA Worldview, Sentinel Hub EO Browser, MetOceanView).

Limitations: The method is limited to raster data and does not yet handle vector fields (e.g., wind or

currents); the dynamic-programming step has quadratic time complexity, which becomes costly beyond roughly 5,000 time steps; and reconstructing the full sequence from a subset of salient frames still loses some spatial continuity.

Research Gap Indicated: Existing satellite/climate data portals expose only a plain timeline or date picker, with no built-in guidance for identifying meaningful events inside long temporal archives.

2.2 Web-Based Visualization and Analytics of Petascale Data: Equity as a Tide that Lifts All Boats

Author(s) / Year: Panta, Huang, McCurdy, Ellsworth, Gooch, Scorzelli, Torres, Klein, Ovando-Montejo, and Pascucci (2024)

Venue: Proceedings of the IEEE Visualization Conference (IEEE VIS)

Objective: To make petabyte-scale NASA climate simulation data explorable in an ordinary web browser, without requiring supercomputing-class hardware or specialised software.

Methodology: The authors introduce a "data fabric" abstraction layer that hides the complexity of the underlying file systems and cloud storage, paired with progressive lossless and lossy compression (ZIP and ZFP) to stream data efficiently over the network. On top of this layer, customisable interactive dashboards are exposed through a simple weblink.

Dataset / Tools: NASA petascale ocean/atmosphere climate simulation output for the Gulf Stream and Kuroshio regions, spanning up to 10,000 time steps across multiple variables.

Key Findings: The dashboards allowed climate scientists and, separately, undergraduate students at a minority-serving institution to interactively explore petabyte-scale datasets from commodity laptops, achieving compression ratios of up to about $3.8\times$ while preserving analytical usefulness.

Limitations: Building and maintaining the abstraction/compression pipeline is engineering-intensive, lossy compression introduces a fidelity–performance trade-off, and the evaluation focuses on ocean/atmosphere simulation output rather than the broader range of imagery types (infrared, radar) used in weather monitoring.

Research Gap Indicated: Earlier climate-visualization tools (e.g., UV-CDAT, CCPviz, Google-Earth-based frameworks) were found to be inefficient at petascale because they required transferring full datasets between architectural layers or downsampling the data, restricting effective use to well-resourced institutions.

2.3 ClimateSERV: An Open Source Earth Observation Climate Data Access Tool

Author(s) / Year: Kruskopf, Gilliland, Roberts, Markert, Ashmall, and Limaye (2025)

Venue: Environmental Modelling & Software

Objective: To lower the barrier for researchers, policymakers and students to visualize, analyse and download Earth-observation-derived climate data for water- and food-security decision-making.

Methodology: A NASA/SERVIR-funded, open-source web application built around extract–transform–load (ETL) pipelines and a public API that automates ingestion, curation and on-demand visualization of curated satellite and model datasets; the tool was integrated into an external drought-forecasting service as a case study.

Dataset / Tools: Curated Earth-observation datasets including CHIRPS precipitation, temperature, soil moisture and vegetation-health indices, accessible through a web GUI and a Python API package.

Key Findings: Over a two-year usage window the platform recorded more than one million data-retrieval and visualization interactions from users across 162 countries, and it was successfully embedded into the Southeast Asia Drought Forecasting Tool.

Limitations: The curated dataset catalogue is oriented toward drought, water and food-security applications rather than the full spectrum of severe-weather imagery (e.g., dedicated cyclone or storm-track layers), and continued operation depends on NASA/SERVIR programme funding.

Research Gap Indicated: Confirms strong, global demand for accessible web tools that bridge complex Earth-observation data formats and non-expert end users, closely mirroring the problem addressed by the assigned platform.

2.4 Comprehensive Analysis of the Use of Web-GIS for Natural Hazard Management: A Systematic Review

Author(s) / Year: Daud, Ugliotti, and Osello (2024)

Venue: Sustainability (MDPI), vol. 16, no. 10, art. 4238

Objective: To synthesise the scattered literature on Web-GIS for natural-hazard management into a coherent picture of state-of-the-art practices, technologies and open issues.

Methodology: A PRISMA-based systematic review that searched SCOPUS and other databases, screening 1,775 articles published between 2014 and 2023 down to 65 studies meeting the inclusion criteria, followed by thematic and geographic analysis of the selected corpus.

Dataset / Tools: Bibliometric records and full texts of the 65 included Web-GIS studies retrieved from SCOPUS and related databases.

Key Findings: Eight recurring sub-themes were identified — spatial analysis and modelling, technology infrastructure, visualization and UI design, decision-support systems, real-time monitoring and early warning, disaster recovery, citizen/social-media integration, and multi-stakeholder collaboration — with research activity concentrated in Italy and China and growing sharply over the last three years.

Limitations: The review is restricted to SCOPUS-indexed, English-language literature from 2014–2023, so it may under-represent grey literature or regionally deployed systems, and it offers a qualitative synthesis rather than a quantitative performance comparison across systems.

Research Gap Indicated: A significant deficit of Web-GIS hazard-management research and deployment was found in low- and middle-income countries, and heatwave monitoring was flagged as an under-served hazard category despite worsening climate conditions.

2.5 A Comprehensive and Up-to-Date Web-Based Interactive 3D Emergency Response and Visualization System Using Cesium Digital Earth: Taking Landslide Disaster as an Example

Author(s) / Year: Yang, Li, Hyypä, Gong, Liu, and Yang (2023)

Venue: Big Earth Data, vol. 7, no. 4, pp. 1058–1080

Objective: To overcome the limited real-time performance, weak interactivity and unrealistic visualization of earlier web-based 3D GIS systems used for disaster emergency response.

Methodology: A full-stack 3D Web-GIS built with the Vue.js front-end framework and the Cesium WebGL API, combining live spatial analysis, interactive route planning, integrated text/image/video messaging, landslide-susceptibility modelling, and progressive 3D reconstruction with augmented-reality visualization using LiDAR and camera input.

Dataset / Tools: 3D terrain models and landslide-susceptibility data demonstrated through a landslide-disaster case study.

Key Findings: The system delivered comprehensive, real-time, interactive 3D visualization and decision support for landslide emergency response, improving situational awareness and inter-agency coordination compared with earlier 2D or static 3D tools.

Limitations: The demonstrated case study covers a single hazard type (landslides), and WebGL-based 3D rendering is resource-intensive, requiring capable client hardware and adequate network bandwidth for smooth real-time interaction.

Research Gap Indicated: Addresses the limited three-dimensional, real-time visualization capability of earlier Web-GIS emergency-response tools, most of which offered only static or near-real-time 2D map views.

2.6 Estimation of Tropical Cyclone Intensity via Deep Learning Techniques from Satellite Cloud Images

Author(s) / Year: Tong, Fu, Deng, Huang, Chan, and He (2023)

Venue: Remote Sensing (MDPI), vol. 15, no. 17, art. 4188

Objective: To estimate tropical cyclone (TC) intensity accurately and efficiently directly from satellite cloud imagery, without relying on hand-crafted feature engineering.

Methodology: Deep convolutional neural networks are trained end-to-end on satellite cloud images (SCIs) to learn intensity-relevant spatial features automatically, replacing the statistical feature-extraction pipelines used in earlier operational techniques.

Dataset / Tools: Historical archives of geostationary infrared satellite cloud imagery for tropical cyclones.

Key Findings: The deep-learning approach achieved improved intensity-estimation accuracy over conventional statistical techniques, demonstrating the practical value of automated feature learning for cyclone monitoring.

Limitations: Model accuracy is closely tied to the quality and resolution of the available cloud imagery and may not generalise well beyond the basin(s) used for training; the approach makes limited use of temporal or sequential information between consecutive frames.

Research Gap Indicated: Confirms that automated, accurate pattern detection for cyclone-related phenomena directly from raw satellite imagery remains an open problem — directly relevant to the anomaly/pattern-detection objective of the assigned platform.

2.7 Estimating Tropical Cyclone Intensity Using an STIA Model From Himawari-8 Satellite Images in the Western North Pacific Basin

Author(s) / Year: Zhang, Liu, Yue, Liu, and Hang (2024)

Venue: IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1–13

Objective: To improve tropical cyclone intensity estimation by explicitly modelling how a cyclone's structure evolves over time, rather than treating each satellite frame in isolation.

Methodology: A Spatiotemporal Interaction Attention (STIA) model combining a spatial-feature-extraction module with temporal-to-spatial and spatial-to-temporal attention modules, applied to sequences of up to 18 consecutive satellite frames.

Dataset / Tools: Himawari-8 geostationary satellite imagery covering tropical cyclones in the western North Pacific basin.

Key Findings: The model achieved a root-mean-square error of 3.61 m/s and a mean absolute error of 2.83 m/s in wind-speed estimation, outperforming prior single-frame and two-branch CNN baselines.

Limitations: Validation is restricted to the western North Pacific basin, so cross-basin generalisability is untested, and processing multi-frame sequences is more computationally demanding than single-frame models.

Research Gap Indicated: Demonstrates the value of temporal-sequence modelling for satellite-based pattern and anomaly detection — a capability not yet incorporated into general-purpose, non-expert-facing visualization platforms such as the one proposed in the assigned problem statement.

3. Comparative Analysis

The table on the following page compares the seven reviewed studies across the research problem addressed, methodology, dataset/tools used, key findings, limitations, and the research gap each study points to. The comparison shows that the reviewed work clusters into three groups: (i) general-purpose climate/satellite data visualization and access platforms [1]–[3]; (ii) Web-GIS architectures for hazard and emergency-response management [4]–[5]; and (iii) AI-based pattern-detection models focused specifically on tropical cyclone intensity [6]–[7]. No single reviewed system combines all three concerns in one accessible, non-expert-facing platform.

Table 1. Comparative Literature Survey

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Chen et al. [1], 2024	Cumbersome, non-intelligent temporal navigation in large geospatial data portals	Need-finding study; autoencoder + dynamic programming; web app (React/D3/Mapbox)	NASA Earthdata, NOAA reanalysis, GOES-16 imagery	Better reconstruction quality than baselines; usability 85.5 vs. 62.5–77.5	Raster only; $O(T^2k)$ DP cost; some continuity loss	No intelligent guidance for temporal selection in current portals
Panta et al. [2], 2024	Petabyte-scale climate data inaccessible without supercomputing hardware	"Data fabric" abstraction layer + progressive compression (ZIP/ZFP); browser dashboards	NASA petascale ocean/atmospheric simulation data (10,000 time steps)	Interactive petabyte-scale analysis from commodity laptops; validated in classroom use	Engineering-intensive pipeline; lossy compression trade-offs	Prior tools inefficient at petascale; restricted to well-resourced users
Kruskopf et al. [3], 2025	Non-experts lack accessible tools to use EO climate data for water/food security	Open-source ETL pipelines + public API; web visualization/analysis/download tool	CHIRPS precipitation, temperature, soil moisture, NDVI	1M+ interactions from 162 countries; integrated into a drought forecasting tool	Catalogue tuned to drought/food security, not general severe weather	Confirms global demand for accessible non-expert EO visualization tools
Daud et al. [4], 2024	Fragmented literature on Web-GIS for natural hazard management	PRISMA systematic review of 1,775 → 65 articles (SCOPUS, 2014–2023)	Bibliometric records from SCOPUS and related databases	8 recurring sub-themes; growth concentrated in Italy and China	SCOPUS-only scope; qualitative synthesis, no benchmarking	Deficit of Web-GIS research/use in low- and middle-income countries
Yang et al. [5], 2023	Earlier 3D Web-GIS emergency-response tools lacked real-time interactivity	Vue.js + Cesium WebGL 3D GIS; route planning, AR/LiDAR reconstruction	3D terrain and landslide-susceptibility data (landslide case study)	Real-time, realistic 3D decision support for emergency response	Single hazard type; resource-intensive WebGL rendering	Limited 3D real-time visualization in earlier Web-GIS tools

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Tong et al. [6], 2023	Efficient, accurate TC intensity estimation from satellite imagery remains hard	End-to-end deep CNNs trained directly on satellite cloud images	Historical geostationary IR satellite cloud image archives	Improved accuracy over conventional statistical techniques	Sensitive to image quality/resolution; little temporal modelling	Automated cyclone pattern detection from raw imagery still open
Zhang et al. [7], 2024	Single-frame models miss the temporal evolution of cyclone structure	Spatiotemporal Interaction Attention (STIA) model on 18-frame sequences	Himawari-8 imagery, western North Pacific basin	RMSE 3.61 m/s, MAE 2.83 m/s; beats single-frame/two-branch CNNs	Single-basin validation; higher compute cost for sequences	Temporal-sequence modelling not yet used in general visualization platforms

4. Critical Analysis and Research Gap

Read together against the assigned problem statement, the reviewed literature reveals five connected gaps.

- Fragmentation of capability. General-purpose climate/satellite access tools such as ClimateSERV [3] and the petascale dashboards of Panta et al. [2] provide broad, accessible data access but little built-in pattern or anomaly detection. Conversely, the cyclone-intensity models of Tong et al. [6] and Zhang et al. [7] achieve strong pattern-detection accuracy but are standalone research models, not embedded in an interactive visualization workflow a non-expert could use directly. No reviewed system integrates ingestion, multi-layer visualization, time-series analytics and automated anomaly detection in a single accessible platform — precisely the integration the assigned problem statement calls for.
- Accessibility versus analytical sophistication. Systems offering the most advanced interaction (SalienTime's salient-frame selection [1], the Cesium 3D emergency-response system of Yang et al. [5]) rely on GPU-accelerated processing, WebGL rendering or custom servers, whereas the most accessible tools (NOAA View, ClimateSERV [3]) offer comparatively shallow analytical depth, such as no automated highlighting of anomalies. A platform aimed at students, small research teams and regional planning bodies must resolve this trade-off rather than choose one side of it.
- Narrow validation scope of pattern-detection models. The tropical-cyclone intensity models reviewed [6], [7] are validated on a single basin or a single hazard type, and their outputs are numerical (wind speed, coordinates) rather than being fed back into an interactive map layer that a decision-maker can inspect visually.
- Geographic and equity gap. The systematic review by Daud et al. [4] found a clear deficit of Web-GIS hazard-management research and deployment in low- and middle-income countries, directly echoing the assigned problem statement's concern for accessibility among "students, small research teams, or regional planning bodies."
- Shallow multi-layer overlay support. Comparative overlay of multiple climate variables (e.g., precipitation, vegetation indices and temperature anomalies) is addressed only partially in the reviewed literature — ClimateSERV [3] and the petascale dashboards [2] support several individual layers, but neither offers a lightweight, browser-only overlay/comparison workflow tuned specifically for disaster-preparedness stakeholders.

5. Future Directions and Relevance to the Assigned Problem Statement

Building on the gaps identified above, the following directions are proposed for the design of the Satellite Data Visualization Platform:

- Adopt an autoencoder-plus-dynamic-programming style salient-time-step mechanism, as in SalienTime [1], so that Objective 3 (time-series analysis tools) can surface meaningful dates within long satellite archives automatically rather than requiring users to scrub through timelines manually.
- Borrow the data-fabric and progressive-compression streaming approach of Panta et al. [2] so the platform's data ingestion pipeline (Objective 1) can serve large NASA/NOAA archives to ordinary browsers and devices without requiring local downloads.
- Embed lightweight CNN/attention-based detection modules, inspired by Tong et al. [6] and Zhang et al. [7], directly into the platform's pattern-detection feature (Objective 4), so that cyclone or temperature anomalies are flagged automatically on the interactive map rather than requiring a separate offline analysis step.

- Apply the Web-GIS design themes identified by Daud et al. [4] — particularly decision-support, real-time monitoring and citizen engagement — to the platform's region-specific filtering (Objective 7) and comparative overlay tool (Objective 5).
- Follow ClimateSERV's [3] model of open ETL pipelines and a public API to keep the platform extensible to new satellite sources and to support the data-export functionality required by Objective 6.
- Treat 3D/WebGL visualization, as demonstrated by Yang et al. [5], as a longer-term enhancement for storm-track and terrain-aware visualization once the core 2D platform (Objective 8: simple web interface) is established, so that added visual sophistication does not compromise accessibility for low-resource users.
- Prioritise deployment choices — browser-only rendering, tiered image resolution, caching — that keep hardware and bandwidth requirements low, directly addressing the equity gap identified in the Web-GIS systematic review [4].

6. Conclusion

The reviewed literature confirms both strong demand for and clear technical building blocks toward an accessible satellite data visualization platform: intelligent temporal navigation [1], scalable web-based access to large climate archives [2], [3], mature Web-GIS design patterns for hazard management [4], [5], and increasingly accurate AI-based pattern detection for severe-weather phenomena [6], [7]. However, no single reviewed system combines these capabilities into one non-expert-facing platform. This gap defines the research opportunity for the assigned Satellite Data Visualization Platform for Weather and Climate Pattern Analysis, and the future directions outlined in Section 5 provide a concrete, literature-grounded path toward addressing it.

References

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- [4] M. Daud, F. M. Ugliotti, and A. Osello, "Comprehensive analysis of the use of Web-GIS for natural hazard management: A systematic review," *Sustainability*, vol. 16, no. 10, art. 4238, May 2024, doi: 10.3390/su16104238.
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- [6] B. Tong, J. Fu, Y. Deng, Y. Huang, P. Chan, and Y. He, "Estimation of tropical cyclone intensity via deep learning techniques from satellite cloud images," *Remote Sensing*, vol. 15, no. 17, art. 4188, 2023, doi: 10.3390/rs15174188.
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